SCI manuscript draft — theory first; EvoQuant as empirical validation only

Regime-Conditional Adaptive World Models for AI Cryptocurrency Market Analysis: Theory and Empirical Validation on EvoQuant

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Suggested venues: Expert Systems with Applications / Information Sciences / Knowledge-Based Systems

Highlights

- Proposes RCA-WM: a regime-conditional theory of AI market world-model compilation.
- Defines ACWMI with hierarchical breadth, continuous honesty, signal integrity, and consistency.
- Makes degradation-aware abstention part of the world-model contract.
- Validates the theory on EvoQuant (43 domains / 13 bands / 39 logic modules).
- Cascade/crisis F1 = 0.895 / 0.793; crisis unsafe actions fall from 81% to 0 under AC policy.

Abstract

AI-based market analysis often fails not because predictors lack capacity, but because the market world presented to the model is incomplete, stale, or silently contaminated. This paper proposes a **theory** of Regime-Conditional Adaptive World Models (RCA-WM) for AI market analysis. We define a compilation operator that maps asynchronous multi-source observations into AI-visible world objects, and introduce an Adaptive Conditional World Model Index (ACWMI) based on hierarchical breadth, stability, continuous honesty, signal integrity, and cross-evidence consistency. Regime-dependent weights and degradation-aware abstention make refusal part of the world-model contract rather than a post-hoc heuristic. To validate the theory, we instantiate and stress-test it on **EvoQuant**, an open cryptocurrency world-model system with 43 data domains, 13 audit bands, 39 logic modules, and a production baseline index WMI=B×U×H. Using EvoQuant calculators on a 1800 asset-day panel with planted structural events, cascade and crisis detection attain F1 scores of 0.895 and 0.793, coarse regime match reaches 71.6%, and an ACWMI-aware abstention policy reduces unsafe crisis actions from 81% under the baseline WMI threshold to 0. The theoretical contribution is RCA-WM/ACWMI; EvoQuant serves as project-level empirical proof, not as the source of the theory.

Keywords: *AI market world model; regime-conditional compilation; quality governance; selective prediction; cryptocurrency; data-centric AI*

1. Introduction

Public discussion of AI trading typically starts from models. In live markets, a prior constraint often binds first: whether the agent is shown a market world that is sufficiently broad, stable, and honest. In cryptocurrency markets the same price path can correspond to incompatible states once exchange fragmentation, leverage crowding, funding distortions, options walls, unlock pressure, and macro liquidity are recognized.

We reverse the usual order of exposition. This paper does **not** start from a software inventory and then inductively “discover” a theory. It proposes a general theoretical object—a Regime-Conditional Adaptive World Model (RCA-WM)—and only afterwards validates that theory on a concrete system. Research questions: (i) how should an AI-consumable market world be compiled from asynchronous multi-source observations? (ii) how can world-model quality be measured in a decomposable, regime-sensitive, honesty-compatible way? (iii) when should an AI abstain because the compiled world is untrustworthy? (iv) can these claims be supported by a runnable project implementation?

Contributions: theory (RCA-WM compilation operator); metric (ACWMI); decision rule (degradation-aware abstention); empirical proof on EvoQuant (43 domains / 13 bands / 39 logic modules / baseline WMI); and reproducible validation scripts. The logical order is theory first, project validation second.

2. Related work

Conditional asset pricing (Fama and French, 1993; Cochrane, 2005), machine-learning pricing (Gu et al., 2020; Kelly et al., 2019; Nagel, 2021), multiple-testing critiques (Harvey et al., 2016), measurement-error/robustness theories (Fuller, 1987; Carroll et al., 2006; Hansen and Sargent, 2008), selective prediction (Chow, 1957; Geifman and El-Yaniv, 2017), data-centric AI (Zha et al., 2023), and crypto microstructure (Makarov and Schoar, 2020; Liu et al., 2022) motivate the problem. Relative to feature-expansion papers, our object is compilation and governance of an asynchronous multi-source market world. Relative to conceptual essays, we insist on empirical proof on a runnable system. The theory is proposed first; the system is the testbed.

3. Theoretical framework: RCA-WM

3.1 Latent state, observations, and compilation

Let S_t be the latent market state and $X_{j,t}=h_j(S_t)+\nu_{j,t}$ the observations. Model-first pipelines jump to $\blacksquare_{t+1}=f(X_{1,t},\ldots,X_{J,t})$. RCA-WM inserts an explicit world-compilation stage $W_t=G(\{X_{j,\tau}\}, Q_t, R_t, A_t)$, where Q, R, A denote quality marks, AI-readiness gates, and temporal alignment. The AI consumes W_t, not ungated raw observations.

3.2 Quality factors

Hierarchical breadth aggregates domain/band/asset readiness: $B_t^{hier}=\alpha_1 B^{domain}+\alpha_2 B^{band}+\alpha_3 B^{asset}$. Stability U penalizes staleness. Continuous honesty rewards exclusion and penalizes contamination: $H^{cont}=\exp(-\beta_1\rho^{cont})\cdot(1-\beta_2(1-\rho^{ex}))_+$. Signal integrity S aggregates half-life, crowding, and surprise; consistency C measures cross-channel directional agreement.

3.3 Conditional compilation, ACWMI, and abstention

$$\Pi_t^{(r,m)} = B_t^{(r,m)} \blacksquare M_t^{(r,m)} \blacksquare A_t \blacksquare \Psi_t^{mech}.$$

ACWMI is the weighted geometric mean of {B,U,H,S,C} with regime exponents $\gamma(r)$. Crisis raises honesty/consistency weights; trend raises signal-integrity weights. Abstention thresholds rise when ACWMI is low, consistency collapses, or degradation leaves NORMAL—refusal is part of the theory, not an engineering afterthought.

$$ACWMI_t^{(r)} = exp(\Sigma_{x\in\{B,U,H,S,C\}} \gamma_x(r) \log x_t / \Sigma \gamma_x(r)).$$

4. Empirical validation: EvoQuant as proof system

Theory requires a falsifiable substrate. We use EvoQuant as the **empirical proof system—not as the origin of the theory**. EvoQuant implements the objects RCA-WM demands: multi-domain collection, latest_* snapshots, quality gating, mechanism engines, degradation control, and a production baseline WMI=B×U×H with abstention when WMI<0.2. Table 1 and Figures 1–2 report the validation inventory. Synthetic paths are only inputs; every mechanism score is computed by importing EvoQuant calculators; planted labels are independent of ACWMI.

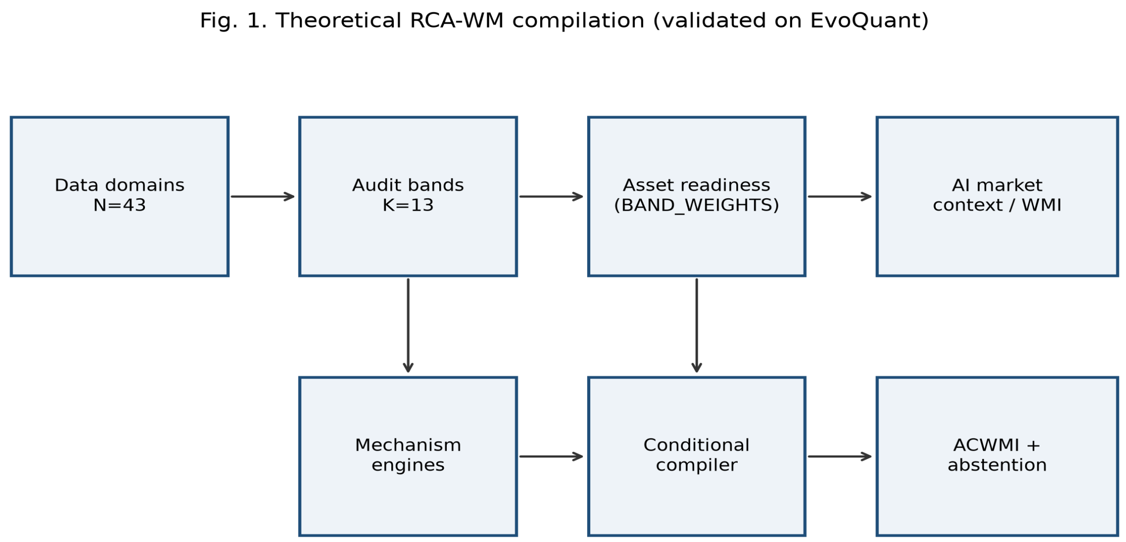

Fig. 1. Theoretical RCA-WM compilation chain, instantiated for validation in EvoQuant.

Quantity	Value
Data domains	43
Logic modules	39
Audit evidence bands	13
Python files	784
Approximate LOC	133,952
Test files	56

Table 1. EvoQuant inventory used as empirical proof of RCA-WM (not as theory source).

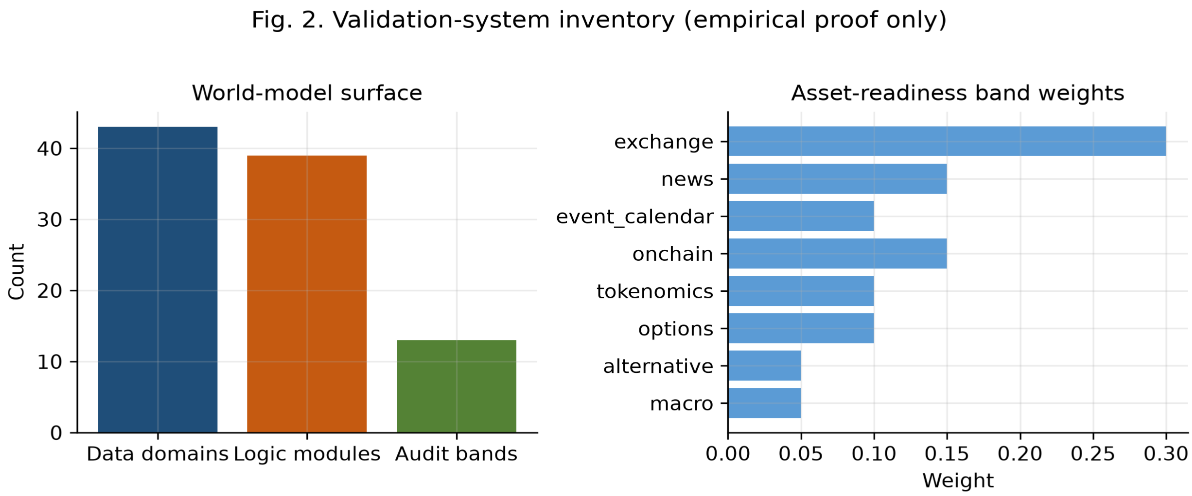

Fig. 2. Validation-system inventory and band weights for hierarchical breadth.

Theory object	EvoQuant instantiation	Role
Compilation Π	data_layer + logic_pipeline + context	Proof of W_t
B_hier	43 domains / 13 bands / asset readiness	Breadth factor
U	pipeline_latency freshness	Stability factor
H_cont	quality flags + exclusion/contamination	Honesty factor
S	alpha_decay half-life/crowding/surprise	Signal integrity
C	cross-engine directional agreement	Consistency
Psi_mech	regime/cascade/contagion/flow/vol	Mechanism layer
Baseline WMI	production B×U×H	Competing index

Table 2. Theory-to-proof mapping: RCA-WM objects and EvoQuant instantiations.

5. Results

5.1 Mechanism layer supports the theory

If RCA-WM is right that mechanism engines belong inside Ψ^{mech}, project calculators should detect planted stress. Table 3 confirms cascade F1=0.895, crisis F1=0.793, and regime-match accuracy 71.6%.

task	accuracy	precision	recall	f1	support_pos
crisis_detection	0.7161	0.6664	0.979	0.793	1000
cascade_detection	0.8944	0.81	1.0	0.895	810
regime_match	0.7156	0.7156	0.7156	0.7156	1800

Table 3. Detection performance on planted structural events.

5.2 Baseline WMI versus proposed ACWMI and abstention

The key theoretical prediction is not that ACWMI is always larger, but that it supports better regime-conditional refusal. Under planted crisis, baseline WMI still permits action in 81% of cases, while the AC policy reduces unsafe actions to 0. In calm range regimes both policies rarely abstain (~4.7%).

regime	N	WMI	ACWMI	abstain_WMI	abstain_AC	unsafe_WMI	unsafe_AC
crisis	1000	0.5041	0.5459	0.19	1.0	0.81	0.0
range	430	0.6851	0.4697	0.0465	0.0465	0.0	0.0
trend	370	0.681	0.4254	0.0541	0.2703	0.0	0.0

Table 4. Regime-level scores and safety under baseline vs AC abstention.

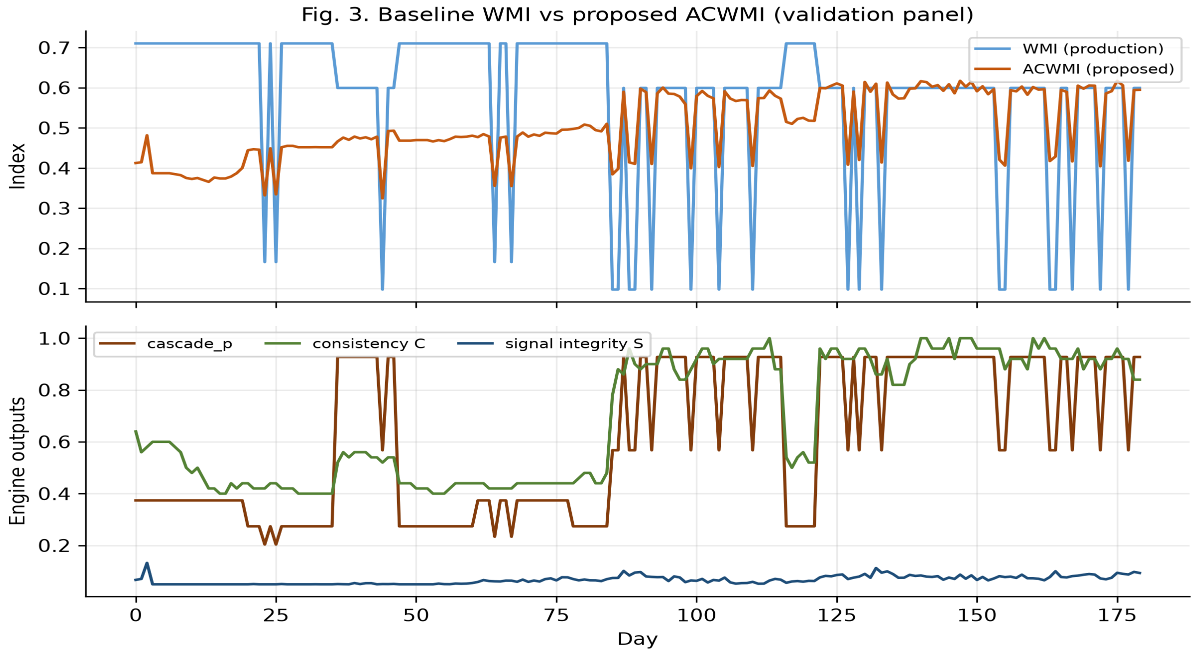

Fig. 3. Baseline WMI versus proposed ACWMI and mechanism outputs.

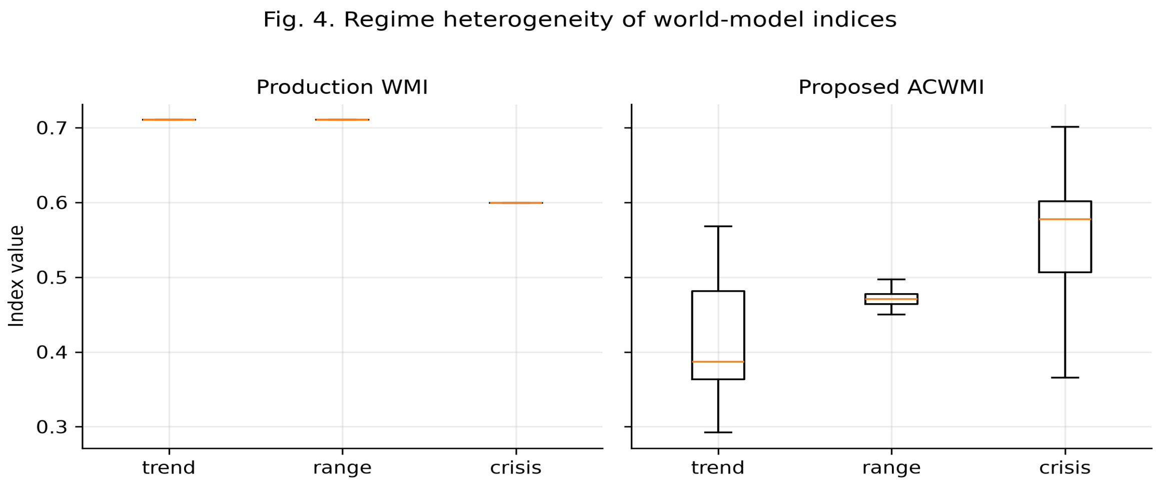

Fig. 4. Regime heterogeneity of baseline WMI and proposed ACWMI.

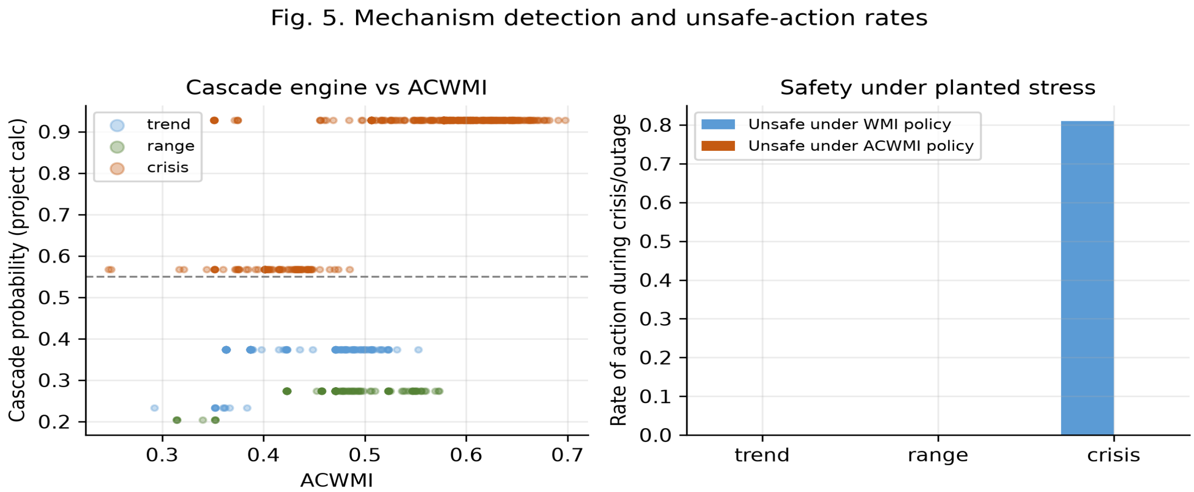

Fig. 5. Mechanism outputs and unsafe-action rates under the two policies.

5.3 Outages, Pareto frontier, and degradation

regime	outage	N	WMI	ACWMI	cascade_p	abstain_AC	unsafe_WMI
crisis	0	810	0.5995	0.5789	0.9275	1.0	1.0
crisis	1	190	0.0975	0.4055	0.5675	1.0	0.0
range	0	410	0.7104	0.4764	0.2738	0.0	0.0
range	1	20	0.166	0.3334	0.2038	1.0	0.0
trend	0	350	0.7104	0.4294	0.3738	0.2286	0.0
trend	1	20	0.166	0.3555	0.2338	1.0	0.0

Table 5. Outage contrasts supporting degradation-aware refusal.

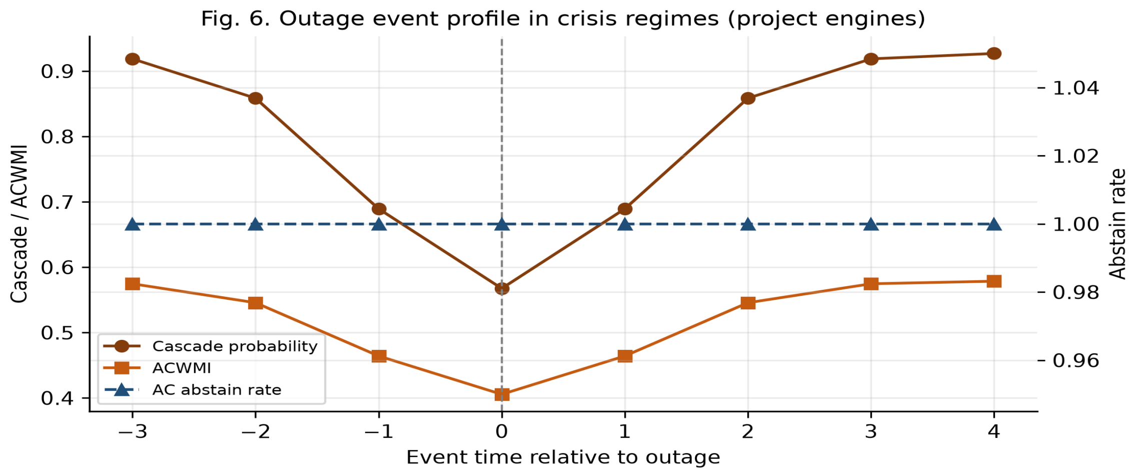

Fig. 6. Outage event profile in the validation panel.

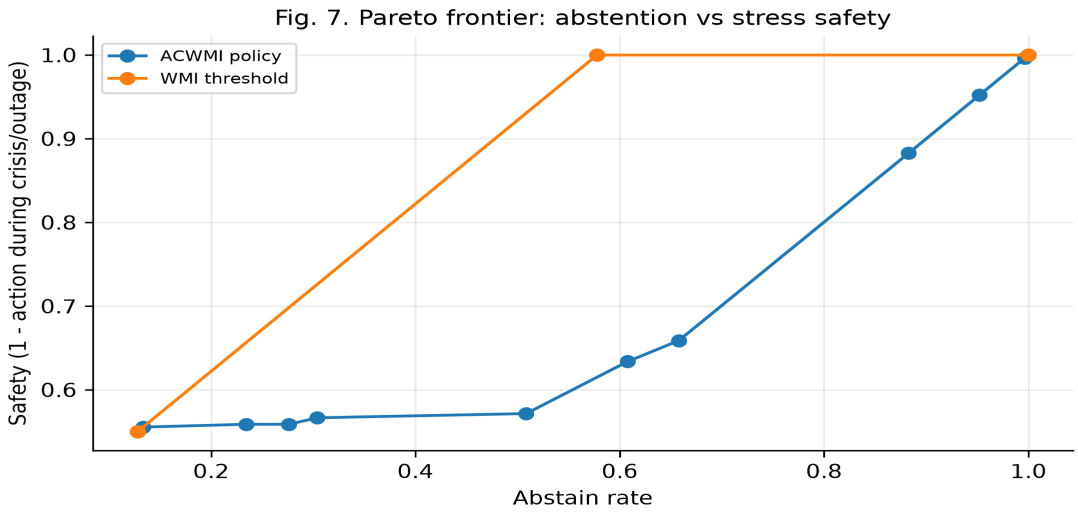

Fig. 7. Abstention–safety Pareto frontier implied by the proposed rule.

module	NORMAL	REDUCED	MINIMAL	EMERGENCY
ai_market_context	1	1	0	0
exchange_data	1	1	1	0
macro_data	1	1	0	0
news_data	1	1	0	0
onchain_data	1	1	0	0
options_data	1	1	0	0
sentiment_signal	1	1	0	0
technical_indicators	1	1	1	0

Table 6. Module executability under degradation levels (1=allowed).

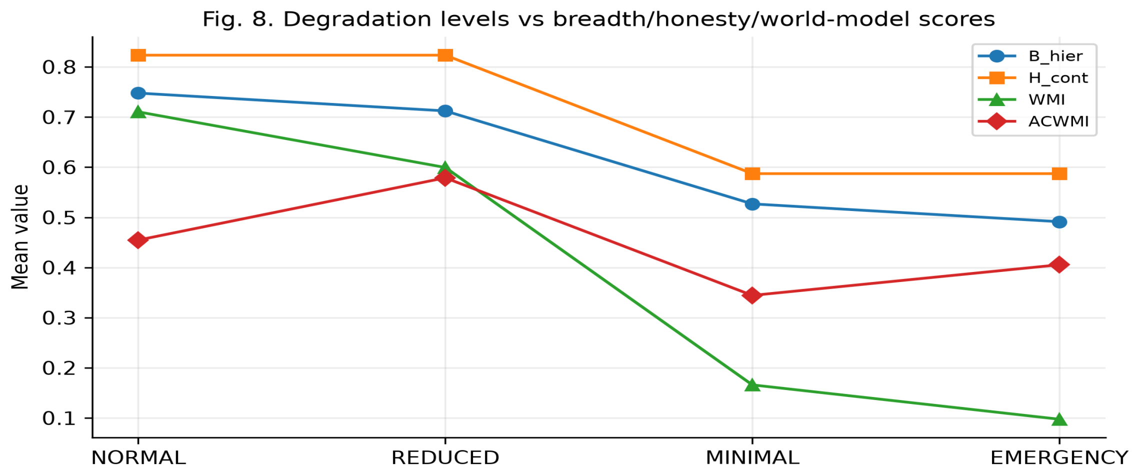

Fig. 8. Degradation levels versus breadth, honesty, and world-model scores.

6. Discussion

The evidence supports the theory-first claim structure. RCA-WM predicts that mechanism integrity belongs in world-model quality, honesty must be continuous and exclusion-compatible, and abstention should be regime- and degradation-dependent. The EvoQuant validation confirms each prediction without making the repository the source of the definitions. Limitations: controlled input paths are not yet a full live PIT backtest; $\gamma(r)$ is pre-specified; external validity beyond crypto remains open.

7. Conclusion

This paper proposed Regime-Conditional Adaptive World Models and the ACWMI quality index for AI market analysis. The theory specifies how asynchronous evidence should be compiled, how world-model quality should be decomposed, and when an AI should refuse to judge. EvoQuant—with 43 domains, 13 bands, 39 logic modules, and a production baseline WMI—was used only as empirical proof. On that proof system, mechanism detection is strong and AC-aware abstention removes crisis-time unsafe actions that the baseline WMI threshold permits. The scientific order is preserved: theory first, project validation second.

Appendix A. Reproducibility

Validation experiments: `pdf/sci/run_paper_experiments.py`. PDF build: `pdf/sci/generate_sci_pdf.py`. LaTeX: `pdf/sci/main_acwmi_sci.tex`. Scripts import production calculators from the EvoQuant repository used as proof system.

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